CS4500 Responsible Computer Science
DSAIT4500 Responsible Data Science and AI Engineering

Group Work Report

Group X

Student A — Student Number
Student B — Student Number
Student C — Student Number
Student D — Student Number

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1 Context

In this report, we address the challenge posed by the discontinuation of the Dutch Crime Anticipation System (CAS) at the end of 2025. CAS predicted crime hotspots on a 125×125 meter grid heatmap based on historical crime data. Despite its ambitions, it was never satisfactorily evidenced to be effective [1], [2]. The Dutch Court of Audit (Algemene Rekenkamer) raised concerns about discriminatory bias and privacy risks, while research showed that human subjectivity inevitably shaped how officers interpreted CAS outputs [3].

We were asked: if the national police approached us for a better technical tool for crime prevention — one that is data-driven yet keeps humans skeptical of automated predictions and in control of final decisions — what would we propose?

The following chapters present our individual ideas (Section 2), subgroup proposals with interaction designs (Section 3), cross-audit feedback and revisions (Section 4), and our generative AI disclosure (Section 5).

2 Individual Thoughts and Questions

2.1 Student A — [Your Name]

2.1.1 Individual Proposal

When I first read about CAS, the thing that stuck out to me was that it was basically just showing where crimes happened before and assuming they'd happen there again. But that doesn't really tell you anything useful. If there's a lot of theft in some area, sure you can send more police there, but that doesn't solve why people are stealing in the first place.

What I'd want to build instead is something that looks at the reasons behind the crime. When you see crime going up somewhere, check what else is happening — are people losing jobs, is it hard to see a doctor, are kids dropping out of school, are utility bills going unpaid? Basically try to connect the crime to the conditions around it.

This would still be a police tool, but it changes what they actually do with the information. Instead of just patrolling an area more, the police would use the system to flag that something deeper is going on and pass it to whoever can actually help — the municipality for infrastructure, GGD for healthcare, social services for support programs. Police can't fix poverty or open a clinic, but they can be the ones to say "this area needs something." That's still crime prevention, just not the patrol-more-cars kind.

I'd also want different people to see different things. An urban planner just sees "this area needs attention." A health department gets the healthcare numbers. The police see that a referral was made but don't get the socioeconomic details — that's not their job. You only see what you can actually do something about. The system doesn't tell anyone what to do, it just goes "hey, look here." Then people figure it out.

2.1.2 Questions for Experts

Question 1 — To someone who actually used CAS in the Dutch police
Expertise: Day-to-day experience with CAS as a police analyst or officer.
Question: What was CAS actually doing for you on a normal workday? Did you find the heatmap helpful or did it mostly just sit there?
Why: I want to know what the real problem was before I try to fix it. What CAS was supposed to do and what it actually did are probably pretty different.

Question 2 — To a legal expert in Dutch law
Expertise: Dutch administrative and criminal law, especially around discrimination.
Question: If a municipality notices socioeconomic indicators going downhill in an area, are they allowed to put extra resources there just based on that? At what point does proactive policy become profiling? Galic, Das, and Schuilenburg (2023) raised similar legal concerns about Dutch predictive systems.
Why: My whole proposal is about acting before things get worse. But I'm not sure if acting on trends rather than actual problems crosses any legal lines.

Question 3 — To a sociologist who studies urban inequality
Expertise: How neighbourhoods get labelled and what happens to people when they do.
Question: If you start directing resources to an area, how do you avoid basically branding it as a problem area? Is there a way to help without making it worse through stigma?
Why: Even well-intentioned flagging could scare off businesses, raise insurance, or make people feel watched. I don't know how to avoid that.

2.2 Student B — [Name]

2.2.1 Individual Proposal

2.2.2 Questions for Experts

2.3 Student C — [Name]

2.3.1 Individual Proposal

2.3.2 Questions for Experts

2.4 Student D — [Name]

2.4.1 Individual Proposal

2.4.2 Questions for Experts

3 Subgroup Proposals

3.1 Subgroup A — [Names]

3.1.1 Tool / Intervention Proposal

We're proposing something we call the **Community Structural Risk Monitor (CSRM)**. The idea is to give the police a way to do crime prevention that goes beyond just showing up more. Instead of a heatmap that says "crime happens here," CSRM tries to figure out *why* it's happening and helps the police coordinate with the right people to deal with it — municipalities, health services, social workers, housing corporations. The police are still running the show, they're just doing something smarter with the information than sending more patrols.

How it works: the system watches two things at the same time. One is regular crime reports from the police — stuff that already happened. The other is what we call stress signals — things like emergency calls, social services referrals, housing complaints, kids dropping out of school. These tend to show up before crime gets worse. When both of these start going up in the same area at the same time, the system opens up a 12-month window and starts pulling in neighbourhood-level data for that area.

The data it collects comes from places like CBS for income statistics, GGD for healthcare access, housing corporations for eviction numbers, and municipal records for things like benefit uptake. None of this is personal data — it's all aggregated at the neighbourhood level. That's a big difference from existing Dutch systems like Top600 in Amsterdam or ProKid 12-SI, which actually create risk scores for individual people [1]. We don't think that's the right approach. Our system looks at the area as a whole and checks if the conditions there match the crime trends. It never singles anyone out.

We also don't just throw in any data we can find. The indicators have to be validated by people who actually know the field — criminologists, sociologists, public health researchers — as stuff that's known to relate to crime. And the most important rule we came up with: you can only see the data if you can actually do something about it. Doctor shortages go to the health ministry. Unemployment numbers go to the labour office. Housing problems go to housing corporations. If you can't fix it, you can't see it. We've been calling this "access follows accountability."

The output is simple. A pin on the map, and only when everything lines up — crime going up, stress signals going up, and the structural indicators matching. No scores, no rankings, no red zones. Just a pin that says "something seems to be going on here, take a look." What happens after that is a human decision.

3.1.2 Human–Machine Interaction Design

The whole point of how CSRM works is that the system helps you notice things, it doesn't decide things for you. That sounds like it should be obvious but CAS showed how quickly it falls apart. Galic, Das and Schuilenburg (2023) wrote about how officers ended up treating the CAS heatmap as if it was telling them what to do, even though it was just a prediction [1], [3]. The map looked authoritative, so people treated it that way. We wanted to avoid that from the start.

3.1.2.1 How the Interaction Works

There are four stages to how someone uses CSRM:

1. **Nothing happens (most of the time).** The system is running in the background, watching the two signal streams across all neighbourhoods. Unless things actually start lining up, it stays quiet.
2. **A pattern shows up.** If the system picks up something — crimes happening in the same spot, same kind of crime, or similar situations for the people involved — it opens a data window and starts gathering the structural indicators for that area.
3. **A pin appears.** If the structural data actually matches the crime and stress pattern over the window period, a pin shows up on the map. That's the only thing the system ever does.
4. **People talk about it.** The pin goes out as a notification to the relevant departments. A panel meets, discusses whether something needs to happen, and decides. The system doesn't get a vote.

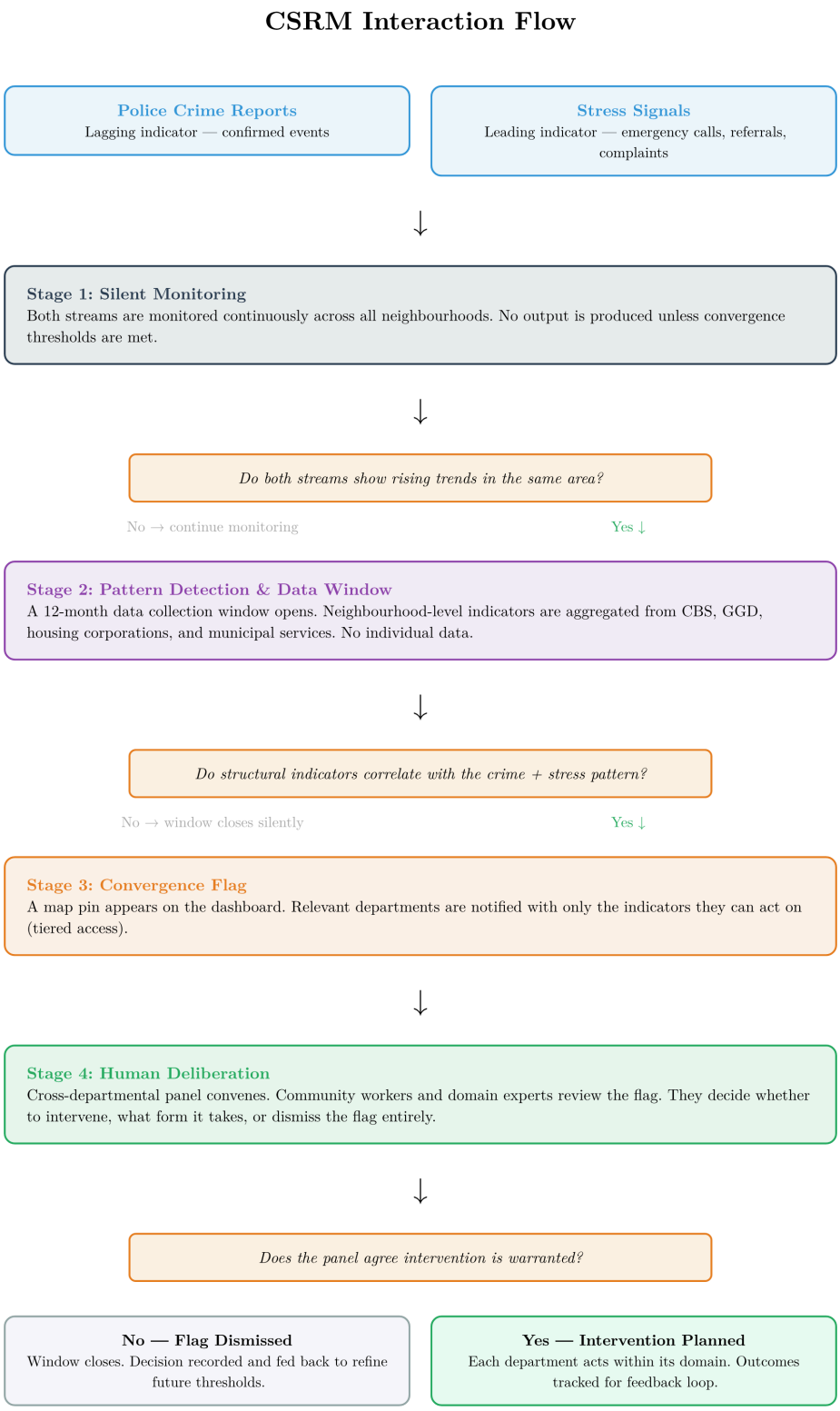

Figure 1: CSRM interaction flow: from monitoring to human deliberation.

3.1.2.2 When People Disagree with the System

We actually expect people to disagree with the flags, and that's fine. Pozzi, Sand, and Jongsma (2026) make the argument for medicine that disagreement between humans and AI isn't a problem to solve, it's actually useful [4]. Same idea here.

A community worker who's been in the neighbourhood for ten years knows stuff that no dataset is going to capture — who actually talks to who, what programs are already running, whether things are actually as bad as the numbers make them look. If they look at a flag and go "no, that's not what's happening," they should be the ones we listen to. The pin is basically asking "did you know about this?" not saying "do something about this."

That's the big difference from CAS. The CAS heatmap felt like a fact [3]. It had colours and squares and it looked like it knew something. Our system is just a dot on a map with a question behind it. It's saying "we noticed something, do you see it too?"

If the experts look at it and decide it's nothing, the window closes and that's the end of it. But that decision gets saved so the validation panel can use it to adjust the thresholds later. So over time the system gets better at knowing when to raise a flag because it's learning from the times humans said "no, you're wrong."

3.1.2.3 Who Sees What

Nobody gets to see everything. Each person only sees what's related to their job:

- **Police** see: that a flag was raised, who got notified, and whether anything came of it. They don't see the income data or the healthcare numbers — that's not what they're there for.
- **Urban planners** see: "this area needs attention." That's it, no details.
- **GGD / health services** see: healthcare stuff only — how long people wait for a GP, whether there's enough addiction support.
- **Labour department** sees: jobs and income data.
- **Housing corporations** see: eviction rates, how much rent is unpaid, building conditions.

This isn't just about privacy for the sake of it. There's a real problem in the Netherlands where data that was collected for one thing ends up being used for something completely different [1], [5]. If the police can't see the socioeconomic data, they can't use it to profile areas. If journalists can't get the flags, there's no "top 10 worst neighbourhoods" list. The way the system is built makes misuse hard, instead of just having a rule that says "please don't misuse this."

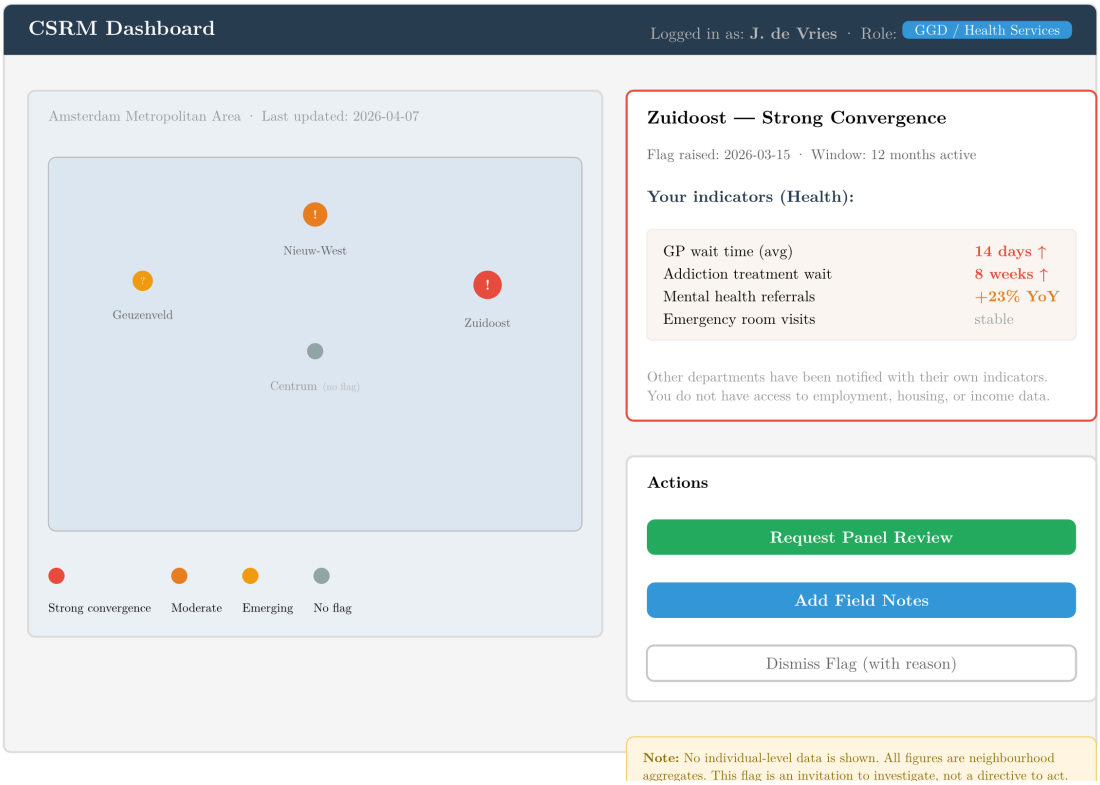

Figure 2: CSRM dashboard mockup — GGD health services view. The logged-in user only sees health-related indicators. Employment, housing, and income data are hidden because that's not their department.

3.2 Subgroup B — [Names]

3.2.1 Tool / Intervention Proposal

3.2.2 Human–Machine Interaction Design

4 Cross-Audit: Subgroup A's Proposal

4.1 Critical Audit by Subgroup B

4.2 Response by Subgroup A

4.3 Revised Proposal by Subgroup A

4.3.1 Revised Tool / Intervention Proposal

4.3.2 Revised Human–Machine Interaction Design

5 Cross-Audit: Subgroup B's Proposal

5.1 Critical Audit by Subgroup A

5.2 Response by Subgroup B

5.3 Revised Proposal by Subgroup B

5.3.1 Revised Tool / Intervention Proposal

5.3.2 Revised Human–Machine Interaction Design

6 Generative AI Disclosure

We used AI during this assignment and are disclosing it here as required.

6.1 Tools and Services Used

- **Claude Code (Anthropic)** — used through the Claude Code CLI as an research assistant and visual designer. Addionaly for organization and documentation. (We used github for version control and Typst for writing, claude was used during the comands and edits of these tools)

6.2 Contributions by Section

6.2.1 Section 2: Individual Thoughts and Questions

Student A: I used Claude as a librarian while developing my proposal. The ideas themselves (looking at causes instead of crime locations, making the police a coordinator rather than the end user, tiered access) came from me. Claude helped me understand some of the terminology — things like ecological correlation and data minimisation — and explained what they actually mean so I could decide if they fit my proposal. I also asked Claude to help me understand the Galic et al. (2023) paper on CAS and Dutch predictive systems, since it's referenced in the assignment and I wanted to know what it actually says before citing it. I had initially a completely diffent idea but existing systems in Dutch police had me change my thoughts.

6.2.2 Section 3: Subgroup Proposals

The technical design came out of the conversation, with me describing how I thought things should work and Claude helping put it into words you see in the diagram. Claude also generated the flow diagram and the dashboard mockup — I described what I wanted them to show and Claude built them in Typst. I also used Claude to look up and understand specific terms and concepts that came up during the design (e.g., what Top600 and ProKid 12-SI actually do, how ecological correlation works in practice, what the Dutch Court of Audit report said about CAS).

Prompts used: "create a mockup and flow diagram based on my proposal." (plus additional manipulation on Typst)

6.3 Motivation and Possible Impacts

We used AI to speed up things we'd have spent a long time on otherwise — reading through papers, figuring out what specific terms mean, and making diagrams. The design decisions and arguments are ours. One downside we ran into is that AI output is usually narrower than what we are trying to achieve in this course. If we hadn't used AI only as an assisntant for our clear purposes and aligned perspective, the report would've sounded like it was written by a different person entirely.

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